

phones where you can get new applications and services with the tap of a button. The software interface harnesses the powerful hardware and gives you a totally new experience. The same would go with respect to cars.

We are racing towards software-defined vehicles as a result of the ongoing transformation of the automobile from a product that is mainly hardware based to a software-centric device on wheels. With the advancements, we are slowly reaching towards better-performing and smart automotive technologies.

As a result, the vehicle cost will increase, and hence leaving that asset unutilised in a parking lot would not be sustainable. This is behind the increasing popularity of shared mobility; the business model of owning vehicles will change. The automotive industry has to adapt to these changes that will happen in various aspects of mobility.

Impact of market trends

With the established market trends such as these, we move towards understanding what they mean. If you look inside a connected electric vehicle, you will observe 10-20 times increase in the software required. The increase is required to cater to those trends. We need to connect with the outside to keep up with the fast changing environment and take actions

instantaneously.

Connectivity also brings the challenge of cyber security. Earlier, the data links were contained inside the vehicle itself, but now everything is going to be exposed on the cloud based infrastructure. Instances like hacking, data manipulation, theft, etc are natural consequences of this connected mobility. It is therefore important to ensure that everything is up to date in the infrastructure and give protection to the users.

With increased complexity in software architecture, need for upgrading the ECUs and other functions, new business models are required where the products will merge. Startups will play an important role here.

Moreover, we are now moving towards improved architecture where we will need around twenty million lines of code for software development. For implementing parallel development, startups will come in to provide that extra fuel for us to go forward.

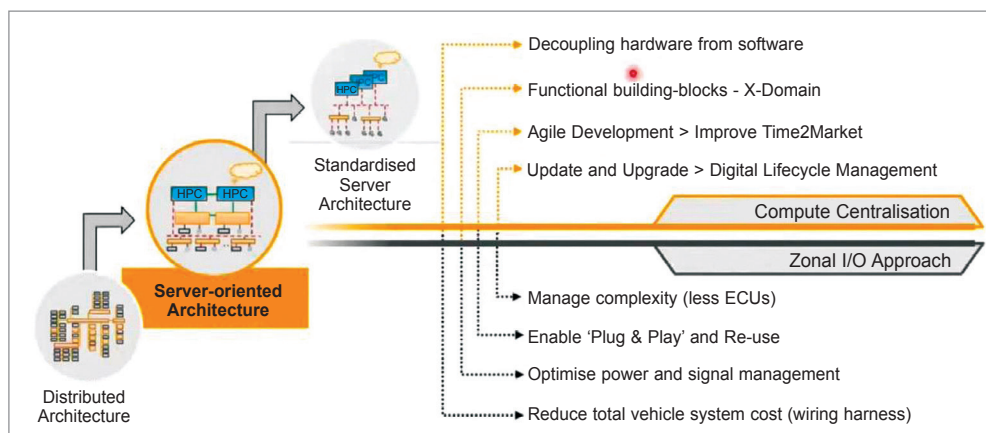


Fig. 3: Architecture trends 2025+

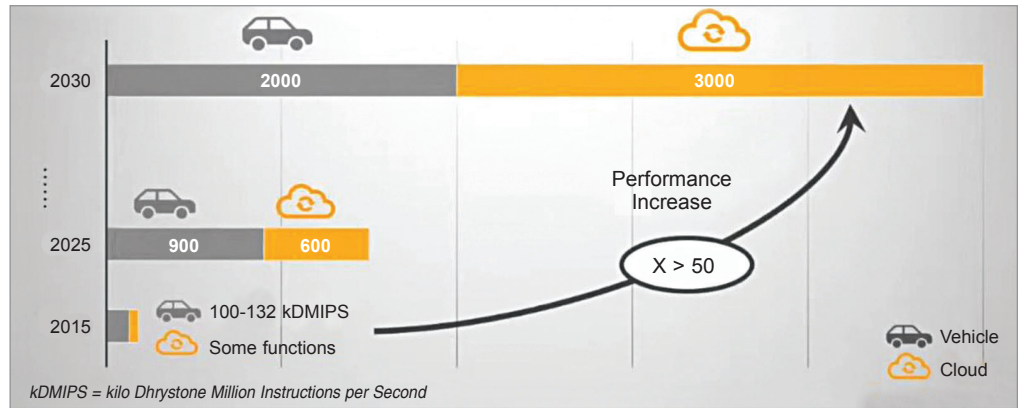


Fig. 2: Computing power in kDMIPS (Source: Based on experience and application of Moore's Law)

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Recent advancements and market trends have made a big impact on vehicle architecture. To cater to all our requirements, vehicles will contain more ECUs, and the computations will increase. Then, with all this movement of data, it is important not to compromise on the real-time reactions that are required. A little compromise can be done on infotainment systems' data responses (not at the cost of user experience), but high priority functions need to be responded to as quickly as possible.

With all these advancements and trends, the cost and the complexity of the vehicle architecture will continue to grow exponentially. Therefore, there is certainly a need to optimise change or upgradation of the vehicle architecture to cater to all these needs.

As cost of the vehicle will be going up, the industry needs to ensure that the vehicle stays relevant.

Architecture trends

In simple terms, there is going to be a big computational revolution. Going forward, we will see a big increase in computational power inside the car, as well as in terms of cloud service. To deliver such computational performance, we are looking at 50 times the performance